AMATH 516: Homework 2
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1. **Exercise 2.2**: Preservation of convexity: Prove the following statements:

   1. Scaling: For any convex set $A \subset \mathbb{E}$, the set $\mathbb{R}_+ A$ is convex.

      Proof. Let $x_1 = \lambda_1 a_1$ and $x_2 = \lambda_2 a_2$ be elements of $\mathbb{R}_+ A$, where $\lambda_1, \lambda_2 \geq 0$ and $a_1, a_2 \in A$. Let $\theta \in [0,1]$. Consider: $\theta x_1 + (1-\theta)x_2 = \theta\lambda_1 a_1 + (1-\theta)\lambda_2 a_2$. Let's further define: $\mu = \theta\lambda_1 + (1-\theta)\lambda_2$. If $\mu = 0$, then both $\lambda_1 = 0$ and $\lambda_2 = 0$, so the combination is 0, which is in $\mathbb{R}_+ A$. Thus, when this combination occurs, we get that $\mathbb{R}_+ A$ is convex. If $\mu > 0$, define:

      $$\begin{aligned} \alpha &= \frac{\theta\lambda_1}{\mu} \\ \beta &= \frac{(1-\theta)\lambda_2}{\mu} \\ \alpha + \beta &= 1 && \text{(From how we defined } \alpha \text{ and } \beta) \\ \theta x_1 + (1-\theta)x_2 &= \mu(\alpha a_1 + \beta a_2) && \text{(Putting it all together)} \end{aligned}$$

      Since A is convex, $\alpha a_1 + \beta a_2 \in A$ and $\mu \geq 0$, so the expression is in $\mathbb{R}_+ A$. Specifically, $\mu(\alpha a_1 + \beta a_2) \in \mathbb{R}_+ A$. We have that $\theta x_1 + (1-\theta)x_2 \in \mathbb{R}_+ A$. Thus, $\mathbb{R}_+ A$ is convex in the nontrivial case as well. So, we have that $\mathbb{R}_+ A$ is convex. □

   2. Set Addition: For any two convex sets $Q_1, Q_2 \subset \mathbb{E}$, the sum $Q_1 + Q_2$ is convex.

      Proof. Let $x, y \in Q_1 + Q2$. By definition of set addition, there exist points $x_1, y_1 \in Q_1$ and $x_2, y_2 \in Q_2$, such that:

      $$x = x_1 + x_2 \text{ and } y = y_1 + y_2$$

      Let $\theta \in [0,1]$. Consider the convex combination:

      $$\begin{aligned} \theta x + (1-\theta)y &= \theta(x_1 + x_2) + (1-\theta)(y_1 + y_2) && \text{(Applying our definitions)} \\ &= (\theta x_1 + (1-\theta)y_1) + (\theta x_2 + (1-\theta)y_2) && \text{(Regrouping terms)} \end{aligned}$$

      Since Q_1 and Q_2 are convex, each set individually satisfies closure under convex combinations that we have applied. Thus, $\theta x_1 + (1-\theta)y_1 \in Q_1$ and $\theta x_2 + (1-\theta)y_2 \in Q_2$. Therefore, $(\theta x_1 + (1-\theta)y_1) + (\theta x_2 + (1-\theta)y_2) \in Q_1 + Q_2$. Since the convex combination of arbitrary elements based on x and y in $Q_1 + Q_2$ remains in $Q_1 + Q_2$, we conclude that $Q_1 + Q_2$ is convex. □

   3. Intersection: The intersection $\bigcap_{i \in I} Q_i$ of convex sets $Q_i \subset \mathbb{E}$, indexed by an arbitrary set I, is convex.

      Proof. Suppose we have an arbitrary indexed family of convex sets $\{Q_i\}_{i \in I}$, where each $Q_i \subset \mathbb{E}$ is convex. Define:

      $$Q = \bigcap_{i \in I} Q_i$$

      To show Q is convex, we must demonstrate closure under convex combinations. Let $x, y \in Q$ such that x and y are chosen arbitrarily in our Q. By definition of intersection, this means: $x, y \in Q_i$ for every $i \in I$. Take an arbitrary scalar $\theta \in [0,1]$. Consider the convex combination: $\theta x + (1-\theta)y$. Since each set Q_i is convex, and we already have $x, y \in Q_i$, we must have, by definition of convexity, $\theta x + (1-\theta)y \in Q_i$ for every $i \in I$. Because this inclusion is true for every

$i \in I$ and $x \in Q_i$ for every $i \in I$ and $y \in Q_i$ for every $i \in I$, it follows directly from the definition of intersection that:

$$\theta x + (1-\theta)y \in \bigcap_{i \in I} Q_i$$

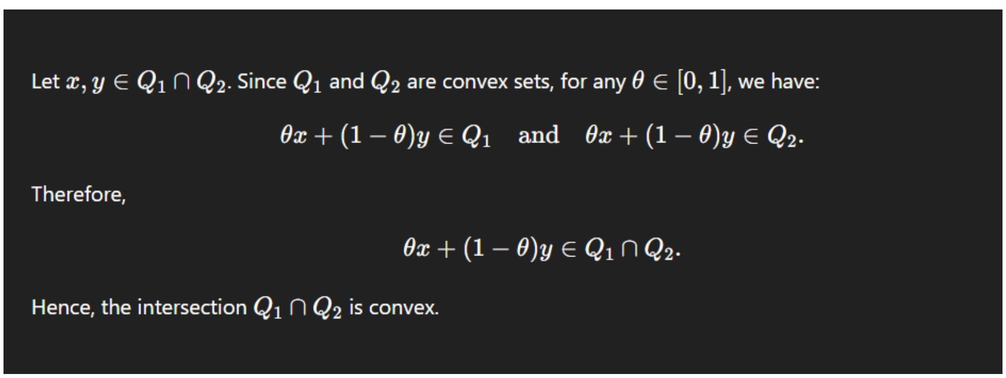
Because x and y were chosen arbitrarily with the condition that $x, y \in Q$ and we have arbitrarily chosen $\theta \in [0,1]$, and resultantly have $(1-\theta) \in [0,1]$, we have shown that the intersection $\bigcap_{i \in I} Q_i$ of convex sets $Q_i \subset \mathbb{E}$, indexed by an arbitrary set I, is convex. ☐

4. Linear Image and Preimage: For any convex sets $Q \subset \mathbb{E}$ and $L \subset \mathbb{Y}$ and a linear map $\mathcal{A} : \mathbb{E} \to \mathbb{Y}$, the image $\mathcal{A}Q$ and the preimage $\mathcal{A}^{-1}L$ are convex sets.

   Proof. Here is the proof for the image:
   Let $y_1, y_2 \in \mathcal{A}Q$. By definition, there exist $x_1, x_2 \in Q$ such that: $y_1 = \mathcal{A}x_1$ and $y_2 = \mathcal{A}x_2$. Let $\theta \in [0,1]$. Consider the statement of convexity: $\theta y_1 + (1-\theta)y_2 = \theta\mathcal{A}x_1 + (1-\theta)\mathcal{A}x_2$. Since $\mathcal{A}$ is linear, we can write: $\theta\mathcal{A}x_1 + (1-\theta)\mathcal{A}x_2 = \mathcal{A}(\theta x_1 + (1-\theta)x_2)$. Since Q is convex, $\theta x_1 + (1-\theta)x_2 \in Q$. Thus, its image under $\mathcal{A}$ is in $\mathcal{A}Q$. Therefore, the image is convex: $\mathcal{A}Q$ is convex. ☐

   Proof. Here is the proof for the preimage:
   Now, consider the preimage $\mathcal{A}^{-1}L$. Let $x_1, x_2 \in \mathcal{A}^{-1}L$. By definition, we have $\mathcal{A}x_1 \in L$ and $\mathcal{A}x_2 \in L$. Let $\theta \in [0,1]$. Consider the combination $\theta\mathcal{A}x_1 + (1-\theta)\mathcal{A}x_2$. That combination is in L and is just a rearrangement for $\mathcal{A}(\theta x_1 + (1-\theta)x_2)$ as we have linearity as a property. Thus, $\theta x_1 + (1-\theta)x_2$ lies in the preimage: $\mathcal{A}^{-1}L$. Hence, the preimage $\mathcal{A}^{-1}L$ is convex. ☐

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:



Let $x, y \in Q_1 \cap Q_2$. Since Q_1 and Q_2 are convex sets, for any $\theta \in [0,1]$, we have:

$$\theta x + (1-\theta)y \in Q_1 \quad \text{and} \quad \theta x + (1-\theta)y \in Q_2.$$

Therefore,

$$\theta x + (1-\theta)y \in Q_1 \cap Q_2.$$

Hence, the intersection $Q_1 \cap Q_2$ is convex.

Here, ChatGPT made a mistake in using Q_1 and Q_2 rather than using the intersection over an indexed set.

2. **Exercise 2.6**: Consider a convex set $Q \subset \mathbb{E}$ and let $k \in \mathbb{N}$ be arbitrary. Show that any convex combination of points $x_1, \ldots, x_k \in Q$ lies in Q.

   Hint: Rewrite $\sum_{i=1}^{k} \lambda_i x_i = (1-\lambda_k) \sum_{i=1}^{k-1} \frac{\lambda_i}{1-\lambda_k} x_i + \lambda_k x_k$ and reason inductively.

   Case $k = 0$: When $k = 0$, we don't really have a combination of points, so we'll state that convexity, here, exists trivially.

   Base Case: $k = 1$:

   Let $x_1 \in Q$ and $\lambda_1 = 1$. Then, $\sum_{i=1}^{1} \lambda_i x_i = x_1$ and remember that $x_1 \in Q$. Because in this combination we only care about $x_1 \in Q$, we have satisfied the Base Case. Because $k = 1$ is where we can start having a combination of points, we will have this as our Base Case.

   Inductive Step:

   For the Inductive Step, we assume that the claim holds for $k = n-1$ with $n \geq 2$. Now, we have that for any $x_1, \ldots, x_{n-1} \in Q$ and nonnegative scalars $\lambda_1, \ldots, \lambda_{n-1}$ satisfying $\sum_{i=1}^{n-1} \lambda_i = 1$, it holds that: $\sum_{i=1}^{n-1} \lambda_i x_i \in Q$. Note that we are not proving that $\sum_{i=1}^{n-1} \lambda_i = 1$, but rather, we are using it as part of our setup for our inductive hypothesis. Now, consider $x_1, \ldots, x_n \in Q$ and scalars $\lambda_1, \ldots, \lambda_n \geq 0$ such that $\sum_{i=1}^{n} \lambda_i = 1$. Let us define μ as $\mu = \sum_{i=1}^{n-1} \lambda_i$.

   Since $\sum_{i=1}^{n} \lambda_i$, we have $\mu + \lambda_n = 1$. We consider two cases: $\mu = 0$ and $\mu > 0$. If $\mu = 0$, then we have that $\lambda_n = 1$ and $\lambda_{i|_{i \neq n}} = 0$. Hence, $\sum_{i=1}^{n} \lambda_i x_i = x_n$ and $x_n \in Q$ meaning that in this convex combination, we have that our convex combination of points trivially lie in Q. We also have the case $\mu > 0$. Here, we define new weights, such that $\alpha_i = \frac{\lambda_i}{\mu}$ for $i = 1, \ldots, n-1$. Note that $\alpha_i \geq 0$ from how we defined λ_i and μ. By the inductive hypothesis, we can define y: $y = \sum_{i=1}^{n-1} \alpha_i x_i$, such that $y \in Q$. Then,

   $$\begin{aligned} \sum_{i=1}^{n} \lambda_i x_i &= (1-\lambda_n) \sum_{i=1}^{n-1} \frac{\lambda_i}{1-\lambda_n} x_i + \lambda_n x_n && \text{(Applying our hint of rewriting)} \\ &= \mu y + \lambda_n x_n && \text{(Applying our definitions of } \mu \text{ and } \lambda) \end{aligned}$$

   As $\mu + \lambda_n = 1$, by convexity of Q and **Lemma 2.4**, we can say $\mu y + \lambda_n x_n \in Q$. For the Inductive Step, we get that $\sum_{i=1}^{n} \lambda_i x_i \in Q$. This completes the inductive step as desired.

   Putting it all together: Thus, by induction, for all $k \in \mathbb{N}$, any convex combination of points $x_1, \ldots, x_k \in Q$ lies in Q.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Inductive Step:

Assume the result holds for $k = n - 1$. That is, any convex combination of $n - 1$ points in Q lies in Q.

Now consider $x_1, \ldots, x_n \in Q$ and nonnegative scalars $\lambda_1, \ldots, \lambda_n$ summing to 1. Define $\mu = \lambda_1 + \cdots + \lambda_{n-1}$.

Then:

- If $\mu = 0$, we have $\lambda_n = 1$, and thus $\sum_{i=1}^{n} \lambda_i x_i = x_n \in Q$.
- If $\mu > 0$, define new coefficients $\alpha_i = \frac{\lambda_i}{\mu}$ for $i = 1, \ldots, n - 1$.

By the inductive hypothesis, $\sum_{i=1}^{n-1} \alpha_i x_i \in Q$.

Thus:

$$\sum_{i=1}^{n} \lambda_i x_i = \mu \left(\sum_{i=1}^{n-1} \alpha_i x_i \right) + (1 - \mu) x_n$$

Since $\mu + (1 - \mu) = 1$, and Q is convex, it follows that the combination $\mu \left(\sum_{i=1}^{n-1} \alpha_i x_i \right) + (1 - \mu) x_n \in Q$.

Hence, by induction, any convex combination of points $x_1, \ldots, x_k$ lies in Q.

Here, ChatGPT made a mistake in defining $\mu = \lambda_1 + \ldots + \lambda_n$ for the Inductive Step. The correct definition of μ should be $\mu = 1 - \lambda_n$ because in the inductive structure, we are separating off the last weight, μ_n, and rewriting the sum accordingly. Defining μ as $1 - \lambda_n$ ensures that the remaining weights $\frac{\lambda_i}{1-\lambda_n}$ for $i = 1, \ldots, n - 1$ form a valid convex combination, which sums to 1 over the first $n - 1$ points. ChatGPT further utilizes μ as if it were $1 - \lambda_n$, but ChatGPT defines $\mu = \lambda_1 + \ldots + \lambda_n$, which is erroneous in nature, and thus, a mistake.

3. **Exercise 2.8**: For any set $Q \subset \mathbb{E}$, prove the equality:

$$\operatorname{conv}(Q) = \left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$$

We will do this proof by using Double Inclusion showing that
$\left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\} \subseteq \operatorname{conv}(Q)$ and
$\operatorname{conv}(Q) \subseteq \left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$.
First to show that $\left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\} \subseteq \operatorname{conv}(Q)$:
Let $x = \sum_{i=1}^{k} \lambda_i x_i$, where $x_i \in Q$ and $\lambda \in \Delta_k$. Since $\operatorname{conv}(Q)$ is a convex set containing Q, it must contain all convex combinations of points in Q by definition. Therefore, $x \in \operatorname{conv}(Q)$. Thus, we see that $\left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\} \subseteq \operatorname{conv}(Q)$.
Second to show that $\operatorname{conv}(Q) \subseteq \left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$:
Observe that $\left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$ is convex.
Let $x = \sum_{i=1}^{k} \lambda_i x_i \in \left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$ and
$y = \sum_{j=1}^{m} \mu_j y_j \in \left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$, where $x_i, y_j \in Q$, $\lambda \in \Delta_k$, and $\mu \in \Delta_m$, and let $\theta \in [0, 1]$. Then, $\theta x + (1-\theta) y = \sum_{i=1}^{k} \theta \lambda_i x_i + \sum_{j=1}^{m} (1-\theta) \mu_j y_j$, which is again a convex combination of elements from Q and the coefficients sum to 1:

$$\sum_{i=1}^{k} \theta \lambda_i + \sum_{j=1}^{m} (1-\theta) \mu_j = \theta + (1-\theta)$$

$$\sum_{i=1}^{k} \theta \lambda_i + \sum_{j=1}^{m} (1-\theta) \mu_j = 1$$

Additionally, for any $x \in Q$, we have that $x = 1 \cdot x$, so $x \in \left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$.
We see that $\left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$ is convex and
$\theta x + (1-\theta) y \in \left\{ \sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k \right\}$. Since $\operatorname{conv}(Q)$ is the intersection of all convex sets containing Q, that means it is actually the smallest convex set containing Q.

Furthermore, $\left\{\sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N}\,,\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k\right\}$ is a convex set containing Q. So, we have $\operatorname{conv}(Q) \subseteq \left\{\sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N}\,,\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k\right\}$.

Thus, we see that $\left\{\sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N}\,,\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k\right\} \subseteq \operatorname{conv}(Q)$ and $\operatorname{conv}(Q) \subseteq \left\{\sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N}\,,\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k\right\}$, we have that by Double Inclusion, $\operatorname{conv}(Q) = \left\{\sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N}\,,\ x_1, \ldots, x_k \in Q\,,\ \lambda \in \Delta_k\right\}$.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

We want to show:

$$\operatorname{conv}(Q) = \left\{\sum_{i=1}^{k} \lambda_i x_i \ : \ k \in \mathbb{N},\ x_1, \ldots, x_k \in Q,\ \lambda \in \Delta_k\right\}.$$

To prove the inclusion $\operatorname{conv}(Q) \subseteq \{\sum_{i=1}^{k} \lambda_i x_i : \ldots\}$, we argue as follows:

Let $C = \left\{\sum_{i=1}^{k} \lambda_i x_i : x_i \in Q,\ \lambda \in \Delta_k,\ k \in \mathbb{N}\right\}$. Then C is convex because it is closed under convex combinations, and clearly $Q \subseteq C$ because any point $x \in Q$ can be written as $x = 1 \cdot x$, which is a convex combination with $k = 1$.

Moreover, since $\operatorname{conv}(Q)$ is defined to be **the union of all convex combinations of finite points from** Q, and C consists precisely of such combinations, we conclude:

$$\operatorname{conv}(Q) \subseteq C.$$

Here, the error or mistake is that ChatGPT defined $\operatorname{conv}(Q)$ to be the union of all convex combinations of finite points from Q rather than the intersection of all convex sets containing Q.

4. **Exercise 2.13**: Show that for any convex set $Q \subset \mathbb{E}$, the two sets, $\operatorname{cl} Q$ and $\operatorname{ri} Q$, are convex and have the same affine hull as Q itself.

   Convexity of $\operatorname{cl} Q$ is Convex:
   Let $x, y \in \operatorname{cl} Q$ and let $\theta \in [0,1]$. We want to show that $\theta x + (1-\theta)y \in \operatorname{cl} Q$. Since $x, y \in \operatorname{cl} Q$, there exist sequences $\{x_k\} \subseteq Q$ and $\{y_k\} \subseteq Q$, such that $x_k \to x$ and $y_k \to y$. Because Q is convex, $\theta x_k + (1-\theta)y_k \in Q$ for all k. The sequence $\theta x_k + (1-\theta)y_k \to \theta x + (1-\theta)y$ and since $Q \subseteq \operatorname{cl} Q$ by definition, we have $\theta x_k + (1-\theta)y_k \in \operatorname{cl} Q$ and the limit also lies in $\operatorname{cl} Q$. Hence, $\theta x + (1-\theta)y \in \operatorname{cl} Q$. So, we see that $\operatorname{cl} Q$ is convex as well.

   Convexity of $\operatorname{ri} Q$ is Convex:
   Let $x, y \in \operatorname{ri} Q$ and let $\theta \in (0,1)$. Since $x, y \in \operatorname{ri} Q$, by definition, there exists $\epsilon > 0$ such that $B_\epsilon(x) \cap \operatorname{aff}(Q) \subseteq Q$ and $B_\epsilon(y) \cap \operatorname{aff}(Q) \subseteq Q$. If $\operatorname{ri} Q$ is an empty set, then convexity is satisfied trivially. So, we continue with the nontrivial case. By **Theorem 2.12**, we know that for any nonempty convex set $Q \subset \mathbb{E}$, the relative interior $\operatorname{ri} Q$ is nonempty. If $x \in \operatorname{ri} Q$, then there exists $\epsilon > 0$ such that $B_\epsilon(x) \cap \operatorname{aff}(A) \subseteq Q$ and if $y \in \operatorname{ri} Q$, then there exists $\epsilon > 0$ such that $B_\epsilon(y) \cap \operatorname{aff}(A) \subseteq Q$. Since $x, y \in \operatorname{ri} Q$, small open balls, which can be called $B_\epsilon(\cdot)$, where $B_\epsilon(\cdot)$ are small open balls, which are chosen arbitrarily, around each point intersected with $\operatorname{aff}(Q)$ lie entirely in Q. Taking convex combinations of points from these balls, or $B_\epsilon(\cdot)$, keeps the result in Q, so any point on the open segment (x, y) must also lie in $\operatorname{ri} Q$. Thus, if $x, y \in \operatorname{ri} Q$, then the entire open line segment $(x, y) \subseteq \operatorname{ri} Q$, and the convex hull of any finite subset of $\operatorname{ri} Q$ is also in $\operatorname{ri} Q$. Hence, $\theta x + (1-\theta)y \in \operatorname{ri} Q$, which proves that $\operatorname{ri} Q$ is convex.

   Showing $\operatorname{aff}(Q) = \operatorname{aff}(\operatorname{cl} Q)$:
   Let $Q \subset \mathbb{E}$. Since $Q \subseteq \operatorname{cl} Q$ by definition, $\operatorname{aff}(Q)$ is the smallest affine set that contains Q. We can use this to see that the affine set is indeed monotonic. Thus, $\operatorname{aff}(Q) \subseteq \operatorname{aff}(\operatorname{cl} Q)$. Note that $\operatorname{cl} Q \subseteq \operatorname{aff}(Q)$ because the closure of Q lies within any closed set containing Q, and $\operatorname{aff}(Q)$ is an affine set containing Q, and also contains its closure meaning that $\operatorname{aff}(\operatorname{cl} Q) \subseteq \operatorname{aff}(Q)$. Because $\operatorname{aff}(Q) \subseteq \operatorname{aff}(\operatorname{cl} Q)$ and $\operatorname{aff}(\operatorname{cl} Q) \subseteq \operatorname{aff}(Q)$, $\operatorname{aff}(Q) = \operatorname{aff}(\operatorname{cl} Q)$.

   Showing $\operatorname{aff}(Q) = \operatorname{aff}(\operatorname{ri} Q)$:
   The affine hull of a set is the smallest affine space that contains a set. Since $\operatorname{ri} Q \subseteq Q$ by definition, $\operatorname{aff}(Q)$ must contain $\operatorname{aff}(\operatorname{ri} Q)$. Thus, $\operatorname{aff}(\operatorname{ri} Q) \subseteq \operatorname{aff}(Q)$. Conversely, since $\operatorname{aff}(\operatorname{ri} Q)$ is an affine space, it must contain Q. Thus, $Q \subseteq \operatorname{aff}(\operatorname{ri} Q)$ meaning $\operatorname{aff}(Q) \subseteq \operatorname{aff}(\operatorname{ri} Q)$ as $\operatorname{aff}(Q)$ is the intersection of all affine sets that contain Q. Because $\operatorname{aff}(\operatorname{ri} Q) \subseteq \operatorname{aff}(Q)$ and $\operatorname{aff}(Q) \subseteq \operatorname{aff}(\operatorname{ri} Q)$, $\operatorname{aff}(Q) = \operatorname{aff}(\operatorname{ri} Q)$.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Affine Hulls:

Since $Q \subseteq \mathrm{cl}(Q)$ and $\mathrm{ri}(Q) \subseteq Q$, it follows by monotonicity that:

$$\mathrm{aff}(Q) \subseteq \mathrm{aff}(\mathrm{cl}(Q)) \quad \text{and} \quad \mathrm{aff}(\mathrm{ri}(Q)) \subseteq \mathrm{aff}(Q).$$

Conversely, since:

$$\mathrm{rb}(Q) = \mathrm{cl}(Q) \cup \mathrm{ri}(Q),$$

we see that $\mathrm{rb}(Q) \supseteq Q$, and therefore:

$$\mathrm{aff}(Q) \subseteq \mathrm{aff}(\mathrm{rb}(Q)).$$

But since $\mathrm{rb}(Q) \subseteq \mathrm{aff}(Q)$, we have:

$$\mathrm{aff}(Q) = \mathrm{aff}(\mathrm{rb}(Q)) = \mathrm{aff}(\mathrm{cl}(Q)) = \mathrm{aff}(\mathrm{ri}(Q)).$$

Here, ChatGPT makes a mistake with the relative boundary by stating that $\mathrm{rb}\, Q = (\mathrm{cl}\, Q) \cup (\mathrm{ri}\, Q)$ when it is actually defined by $\mathrm{rb}\, Q = (\mathrm{cl}\, Q) \setminus (\mathrm{ri}\, Q)$.

5. **Exercise 2.16**: Show that for any nonempty set $Q \subseteq \mathbb{E}$, the function $\mathrm{dist}_Q : \mathbb{E} \to \mathbb{R}$ is 1 Lipschitz.

   First, we use and apply the basic definition of distance function: $\mathrm{dist}_Q(x) = \inf_{y \in Q} \|x-y\|$ and $\mathrm{dist}_Q(z) = \inf_{y \in Q} \|z-y\|$.

   We apply Triangle Inequality: For any $y \in Q$, we have $\|x-y\| \le \|x-z\| + \|z-y\|$.
   Therefore, $\inf_{y \in Q} \|x-y\| \le \inf_{y \in Q}(\|x-z\| + \|z-y\|)$ meaning that $\mathrm{dist}_Q \le \inf_{y \in Q}(\|x-z\| + \|z-y\|)$.
   Since $\|x-z\|$ does not depend on y, we have $\mathrm{dist}_Q(x) \le \|x-z\| + \inf_{y \in Q} \|z-y\|$.
   Thus, $\mathrm{dist}_Q(x) \le \|x-z\| + \mathrm{dist}_Q(z)$ meaning that $\mathrm{dist}_Q(x) - \mathrm{dist}_Q(z) \le \|x-z\|$.

   We apply Triangle Inequality: For any $y \in Q$, we have $\|z-y\| \le \|z-x\| + \|x-y\|$.
   Therefore, $\inf_{y \in Q} \|z-y\| \le \inf_{y \in Q}(\|z-x\| + \|x-y\|)$ meaning that $\mathrm{dist}_Q \le \inf_{y \in Q}(\|z-x\| + \|x-y\|)$.
   Since $\|z-x\|$ does not depend on y, we have $\mathrm{dist}_Q(z) \le \|z-x\| + \mathrm{dist}_Q(x)$.
   Thus, $\mathrm{dist}_Q(z) \le \|z-x\| + \mathrm{dist}_Q(x)$ meaning that $\mathrm{dist}_Q(z) - \mathrm{dist}_Q(x) \le \|z-x\|$.

   We have $\mathrm{dist}_Q(x) - \mathrm{dist}_Q(z) \le \|x-z\|$ and $\mathrm{dist}_Q(z) - \mathrm{dist}_Q(x) \le \|z-x\|$. Thus, $\|\mathrm{dist}_Q(x) - \mathrm{dist}_Q(z)\| \le \|x-z\|$ and $\|\mathrm{dist}_Q(z) - \mathrm{dist}_Q(x)\| \le \|z-x\|$. We see that by definition of Lipschitz continuity and what it means to be 1 Lipschitz, the function $\mathrm{dist}_Q : \mathbb{E} \to \mathbb{R}$ is 1 Lipschitz.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Proof:

Let $x, z \in \mathbb{E}$. First, apply the triangle inequality: for any $y \in Q$,

$$\|x - y\| \leq \|x - z\| + \|z - y\|.$$

Taking the infimum over all $y \in Q$, we get:

$$\text{dist}_Q(x) \leq \|x - z\| + \text{dist}_Q(z).$$

Thus,

$$\text{dist}_Q(x) - \text{dist}_Q(z) \leq \|x - z\|.$$

Now switch roles: again using the triangle inequality,

$$\|z - y\| \leq \|z - x\| + \|x - y\|.$$

We now note that $\|z - x\| = -\|x - z\|$, so:

$$\|z - y\| \leq -\|x - z\| + \|x - y\|.$$

Taking the infimum over all $y \in Q$, this implies:

$$\text{dist}_Q(z) \leq -\|x - z\| + \text{dist}_Q(x),$$

and rearranging gives:

$$\text{dist}_Q(z) - \text{dist}_Q(x) \leq -\|x - z\|.$$

Combining the two inequalities, we find:

$$|\text{dist}_Q(x) - \text{dist}_Q(z)| \leq \|x - z\|,$$

so dist_Q is 1-Lipschitz.

Here, ChatGPT makes an error by stating $\|z - x\| = -\|x - z\|$ when indeed $\|z - x\| \neq -\|x - z\|$ as norms are strictly nonnegative in nature. Instead, $\|z - x\| = \|x - z\|$.

6. **Exercise 2.18**: Show that for any nonempty, closed convex set $Q \subset \mathbb{E}$, the map $x \mapsto \text{proj}_Q(x)$ is 1 Lipschitz.

   Let's choose x_1, x_2 arbitrarily such that $x_1, x_2 \in \mathbb{E}$. Furthermore, let $p = \text{proj}_Q(x_1)$ and $q = \text{proj}_Q(x_2)$. Applying the properties of projection, we have:

$$\langle x_1 - p, q - p \rangle \leq 0 \qquad \text{(Equation 1)}$$

$$\langle x_2 - q, p - q \rangle \leq 0 \qquad \text{(Equation 2)}$$

   Now, add Equation 1 and Equation 2 together to get: $\langle x_1 - p, q - p \rangle + \langle x_2 - q, p - q \rangle \leq 0$. By using properties of inner products, we can rewrite this to say: $\langle x_1 - p, q - p \rangle - \langle x_2 - q, q - p \rangle \leq 0$. So, we have $\langle (x_1 - p) - (x_2 - q), q - p \rangle \leq 0$, which means $\langle x_1 - p - x_2 + q, q - p \rangle \leq 0$. Rearranging accordingly, we have $\langle (x_1 - x_2) + (q - p), q - p \rangle \leq 0$. We will split the inner product now: $\langle x_1 - x_2, q - p \rangle + \langle q - p, q - p \rangle \leq 0$. And from this we will see, $\langle x_1 - x_2, q - p \rangle + \|q - p\|^2 \leq 0$ as $\langle q - p, q - p \rangle = \|q - p\|^2$. This rearranges to $\langle x_1 - x_2, p - q \rangle \geq \|p - q\|^2$. We now apply the Cauchy-Schwarz inequality, giving us $|\langle x_1 - x_2, p - q \rangle| \leq \|x_1 - x_2\| \cdot \|p - q\|$, which leads to $\|x_1 - x_2\| \geq \|p - q\|$. Remember that $p = \text{proj}_Q(x_1)$ and $q = \text{proj}_Q(x_2)$ meaning that $\|x_1 - x_2\| \geq \|\text{proj}_Q(x_1) - \text{proj}_Q(x_2)\|$ and rewriting the inequality $\|\text{proj}_Q(x_1) - \text{proj}_Q(x_2)\| \leq \|x_1 - x_2\|$. Because $\|\text{proj}_Q(x_1) - \text{proj}_Q(x_2)\| \leq \|x_1 - x_2\|$, we have that $x \mapsto \text{proj}_Q(x)$ is 1 Lipschitz as desired.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Let $x_1, x_2 \in \mathbb{E}$, and let $p = \mathrm{proj}_Q(x_1)$, $q = \mathrm{proj}_Q(x_2)$, where $Q \subseteq \mathbb{E}$ is nonempty, closed, and convex. By properties of projections, we have:

$$\langle x_1 - p, q - p \rangle \leq 0 \quad (1)$$

$$\langle x_2 - q, p - q \rangle \leq 0 \quad (2)$$

Adding (1) and (2), we get:

$$\langle x_1 - p, q - p \rangle + \langle x_2 - q, p - q \rangle \leq 0$$

Using properties of inner products and simplifying:

$$\langle (x_1 - x_2) + (q - p), q - p \rangle \leq 0$$

This expands to:

$$\langle x_1 - x_2, q - p \rangle + \|q - p\|^2 \leq 0$$

Rewriting this:

$$\langle x_1 - x_2, q - p \rangle \leq -\|q - p\|^2$$

Now we apply Cauchy-Schwarz:

$$|\langle x_1 - x_2, q - p \rangle| \leq \|x_1 - x_2\| \cdot \|q - p\|$$

So:

$$-\|q - p\|^2 \leq \|x_1 - x_2\| \cdot \|q - p\|$$

Then we divide both sides by $\|q - p\|$ and use the fact that $-\|p - q\| = \|q - p\|$, giving us:

$$\|q - p\| \leq \|x_1 - x_2\|$$

Hence, the projection map is 1-Lipschitz.

Here, ChatGPT makes an error by stating $-|p - q\| = \|q - p\|$ when indeed $-|p - q\| \neq \|q - p\|$ as norms are strictly nonnegative in nature. Instead, $\|p - q\| = \|q - p\|$.

7. **Exercise 2.23**: Show that a set $K \subset \mathbb{E}$ is a convex cone if and only if the point $\lambda x + \mu y$ lies in K for any two points $x, y \in K$ and numbers $\lambda, \mu \geq 0$.

   We will do this Proof by Biconditional:

   First, we show that if a set $K \subset \mathbb{E}$ is a convex cone, then the point $\lambda x + \mu y$ lies in K for any two points $x, y \in K$ and numbers $\lambda, \mu \geq 0$. A convex cone, by definition, is closed under addition and positive scalar multiplication. If $x \in K$, then for any $\lambda \geq 0$, $\lambda x \in K$, and if $y \in K$, then for any $\mu \geq 0$, $\mu y \in K$. Since K is closed under addition, because $\lambda x \in K$ and $\mu y \in K$, $\lambda x + \mu y \in K$ as $x, y \in K$ defined by the problem statement. Therefore, for any $x, y \in K$ and $\lambda, \mu \geq 0$, $\lambda x + \mu y \in K$ as desired.

   Second, we show that if the point $\lambda x + \mu y$ lies in K for any two points $x, y \in K$ and numbers $\lambda, \mu \geq 0$, then the set $K \subset \mathbb{E}$ is a convex cone. We want to show that K is a convex cone. First, we verify that K is a cone. Take any $x \in K$ and any scalar $\alpha \geq 0$. By setting $y = x$ and $\mu = 0$, the assumption gives $\alpha x + 0 \cdot y = \alpha x$, where $\alpha x \in K$, so K is closed under nonnegative scalar multiplication. Next, we verify convexity. Take any $x, y \in K$ and let $\theta \in [0, 1]$. Setting $\lambda = \theta$ and $\mu = 1 - \theta$, both of which are nonnegative, we have $\theta x + (1 - \theta) y \in K$. Thus, K is closed under convex combinations and hence convex. Therefore, K is both a cone and convex meaning that K is a convex cone.

   Thus, because we have shown if a set $K \subset \mathbb{E}$ is a convex cone, then the point $\lambda x + \mu y$ lies in K for any two points $x, y \in K$ and numbers $\lambda, \mu \geq 0$ and if the point $\lambda x + \mu y$ lies in K for any two points $x, y \in K$ and numbers $\lambda, \mu \geq 0$, then the set $K \subset \mathbb{E}$ is a convex cone, we have that a set $K \subset \mathbb{E}$ is a convex cone if and only if the point $\lambda x + \mu y$ lies in K for any two points $x, y \in K$ and numbers $\lambda, \mu \geq 0$.

   Acknowledgment: For this question, I got help from Collin.

   Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

   > We want to show that a set $K \subset \mathbb{E}$ is a convex cone if and only if for all $x, y \in K$ and $\lambda, \mu \geq 0$, the point $\lambda x + \mu y \in K$.
   >
   > **(⟹)** Assume K is a convex cone. Then for any $x, y \in K$ and $\lambda, \mu \geq 0$, clearly $x + y \in K$ since convexity says any sum lies in the set. Also, since $\lambda, \mu \geq 0$, we have $\lambda x + \mu y \in K$ because cones are closed under scalar addition.
   >
   > **(⟸)** Now suppose that $\lambda x + \mu y \in K$ for all $x, y \in K$ and $\lambda, \mu > 0$. We want to show that K is a convex cone. First, for any $x \in K$ and $\alpha > 0$, we can set $y = 0$ and $\mu = 0$, so $\lambda x + \mu y = \alpha x + 0 = \alpha x \in K$. So K is a cone.
   >
   > Next, to show convexity, we pick $\lambda = \mu = \frac{1}{2}$ and get $\frac{1}{2}x + \frac{1}{2}y \in K$. So, since we showed it for one pair of weights, K must be convex.
   >
   > Therefore, K is a convex cone.

   Here, the error is the assumption that $x + y \in K$. This isn't necessarily true and is a mistake.

8. **Exercise 2.27**: Prove for any nonempty cone $K \subset \mathbb{E}$ the equality: $(K^\circ)^\circ = \operatorname{cl}\operatorname{conv}(K)$.

   We know that by definition:

   $$K^\circ = \{v \in \mathbb{E} : \langle v, x\rangle \leq 0 \text{ for all } x \in K\}$$

   Thus, K° consists of all vectors v that make an obtuse angle with every vector $x \in K$.
   Now, we apply this definition again:

   $$(K^\circ)^\circ = \{v \in \mathbb{E} : \langle v, x\rangle \leq 0 \text{ for all } x \in K^\circ\}$$

   Thus, $(K^\circ)^\circ$ consists of all vectors v that make an obtuse angle with every vector $x \in K^\circ$.

   We will proceed with this proof by Double Inclusion.

   In order to show that $\operatorname{cl}\operatorname{conv}(K) \subseteq (K^\circ)^\circ$, we let $z \in \operatorname{cl}\operatorname{conv}(K)$. Then, there exists a sequence $z_n \in \operatorname{conv}(K)$ such that $z_n \to z$. Take any $v \in K^\circ$. By definition, $\langle v, x\rangle \leq 0$ for all $x \in K$. Then, since $z_n \in \operatorname{conv}(K)$, we have $\langle v, z_n\rangle \leq 0$ for all n. Because $z_n \to z$, we can say $\langle v, z\rangle \leq 0$. Hence, $z \in (K^\circ)^\circ$. So, we can conclude $\operatorname{cl}\operatorname{conv}(K) \subseteq (K^\circ)^\circ$.

   In order to show that $(K^\circ)^\circ \subseteq \operatorname{cl}\operatorname{conv}(K)$, we recall that: $(K^\circ)^\circ = \{v \in \mathbb{E} : \langle v, x\rangle \leq 0 \text{ for all } x \in K^\circ\}$. So, we will show that $\{v \in \mathbb{E} : \langle v, x\rangle \leq 0 \text{ for all } x \in K^\circ\} \subseteq \operatorname{cl}\operatorname{conv}(K)$. In other words, we will have to show that all vectors v that make an obtuse angle with every vector $x \in K^\circ$ are actually in $\operatorname{cl}\operatorname{conv}(K)$. We know that cones contain the origin as we can set $\lambda = 0$ in definition of a cone. So, now, we will proceed with Proof by Contradiction. We will take a vector $v \in (K^\circ)^\circ$, where $v \notin \operatorname{cl}\operatorname{conv}(K)$. Since $\operatorname{cl}\operatorname{conv}(K) \subset \mathbb{E}$ is a nonempty, closed, convex set and $v \notin \operatorname{cl}\operatorname{conv}(K)$, we may apply Strict Separation. Then, there exists a nonzero vector $a \in \mathbb{E}$ and a scalar $b \in \mathbb{R}$ such that $\langle a, x\rangle \leq b < \langle a, v\rangle$ for all $x \in \operatorname{cl}\operatorname{conv}(K)$. Now, we plug in $x = 0$. Since $0 \in \operatorname{cl}\operatorname{conv}(K)$, it follows that

   $$\langle a, 0\rangle = 0$$

   $$0 \leq b$$

   $$b \geq 0$$

   Thus, we see that b is nonnegative.
   We also see from strict separation $b < \langle a, v\rangle$, so $\langle a, v\rangle > 0$ as well.
   We note that K is a cone and homogeneous by properties of a cone.
   If $x \in K$, then for every $\lambda \geq 0$, we also have $\lambda x \in K$ by properties of a cone. Hence, for all $x \in K$, we have that the ray generated by x is contained entirely within K. Now, we have that $\langle a, x\rangle \leq b$ for all $x \in \operatorname{cl}\operatorname{conv}(K)$. Because K is a cone, we can particularly examine what happens if we scale vectors in K.

   $$\langle a, x\rangle \leq b$$

   $$\langle a, \lambda x\rangle \leq b \qquad \text{(Applying homogeneity)}$$

   $$\langle a, x\rangle \leq \frac{b}{\lambda} \qquad \text{(Dividing through by } \lambda\text{)}$$

   $$\langle a, x\rangle \leq \lim_{\lambda\to\infty} \frac{b}{\lambda}$$

   $$\langle a, x\rangle \leq 0$$

   We see that every x in the cone satisfies this inequality.
   By definition, we see that $a \in K^\circ$. Recall that $v \in (K^\circ)^\circ$. By Strict Separation, $\langle a, v\rangle > 0$. Because $a \in K^\circ$ and $v \in (K^\circ)^\circ$, we see that an obtuse angle is made meaning that by definition, $\langle a, v\rangle \leq 0$,

which leads to a contradiction. This means that $v \in \operatorname{cl}\operatorname{conv}(K)$. Thus, we see that $(K^\circ)^\circ \subseteq \operatorname{cl}\operatorname{conv}(K)$ as v was chosen arbitrarily such that $v \in (K^\circ)^\circ$.

Because $\operatorname{cl}\operatorname{conv}(K) \subseteq (K^\circ)^\circ$ and $(K^\circ)^\circ \subseteq \operatorname{cl}\operatorname{conv}(K)$, we have that $(K^\circ)^\circ = \operatorname{cl}\operatorname{conv}(K)$.

Acknowledgment: For this question, I got help from Collin. Additionally, I referenced Variational Analysis by R T Rockafellar.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

> We aim to show that $(K^\circ)^\circ \subseteq \mathbf{cl}(\mathbf{conv}(K))$. Suppose $v \in (K^\circ)^\circ$ but $v \notin \mathbf{cl}(\mathbf{conv}(K))$. Then by the Strict Separation Theorem, there exists a nonzero vector $a \in \mathbb{E}$ and a scalar $b \in \mathbb{R}$ such that:
>
> $$\langle a, x\rangle \leq b < \langle a, v\rangle \quad \text{for all } x \in \mathbf{cl}(\mathbf{conv}(K)).$$
>
> Now since all cones are closed sets by definition, we know $K = \mathbf{cl}(K)$, so $\mathbf{cl}(\mathbf{conv}(K)) = \mathbf{conv}(K)$. Therefore, $v \notin \mathbf{conv}(K)$, and we can write v as a linear combination of vectors outside of K.

Not all cones are closed sets. ChatGPT makes a mistake here.

9. **Exercise 2.29**: Show that for any set $Q \subset \mathbb{E}$ containing the origin, equality holds:

$$Q^\circ = \{v \in \mathbb{E} : \langle v, x\rangle \leq 1 \text{ for all } x \in Q\}$$

We just apply the definition of polarity of a cone

$$\begin{aligned} K^\circ &= \{v \in \mathbb{E} : \langle v, x\rangle \leq 0 \text{ for all } x \in K\} \\ &= \{(v, \alpha) \in \mathbb{E} \times \mathbb{R} : \langle (v, \alpha), (\lambda x, \lambda)\rangle \leq 0 \text{ for all } (\lambda x, \lambda) \in K\} \\ &= \{(v, \alpha) \in \mathbb{E} \times \mathbb{R} : \lambda(\langle v, x\rangle + \alpha) \leq 0 \text{ for every } x \in Q\,,\ \lambda \geq 0\} \end{aligned}$$

Consider $v \in \mathbb{E}$ such that $(v, -1) \in K^\circ$. This would mean that $\lambda\langle v, x\rangle - \lambda \leq 0$ for all $x \in Q$, where $\lambda \geq 0$. Equivalently, $\langle v, x\rangle \leq 1$ for all $x \in Q$. Thus, $Q^\circ \subseteq \{v \in \mathbb{E} : \langle v, x\rangle \leq 1 \text{ for all } x \in Q\}$.

Now, consider $v \in \mathbb{E}$ such that $\langle v, x\rangle \leq 1$ for all $x \in Q$:

$$\forall x \in Q\,,\ \langle v, x\rangle \leq 1$$

$$\forall x \in Q\,,\ \forall \lambda \geq 0\,,\ \lambda(\langle v, x\rangle - 1) \leq 0$$

$$(v, -1) \in K^\circ$$

$$v \in Q^\circ$$

Now, we can see: $\{v \in \mathbb{E} : \langle v, x\rangle \leq 1 \text{ for all } x \in Q\} \subseteq Q^\circ$.

Because $Q^\circ \subseteq \{v \in \mathbb{E} : \langle v, x\rangle \leq 1 \text{ for all } x \in Q\}$ and $\{v \in \mathbb{E} : \langle v, x\rangle \leq 1 \text{ for all } x \in Q\} \subseteq Q^\circ$, we have that $Q^\circ = \{v \in \mathbb{E} : \langle v, x\rangle \leq 1 \text{ for all } x \in Q\}$.

Acknowledgment: For this question, I got help from Shawn.

Extra Credit: After completing the exercise, I asked Claude AI to do the question and this is what it stated:

> S Show that for any set $Q \subset \mathbb{E}$ containing the origin, equality holds: $$Q^\circ=\{v \in \mathbb{E}: \langle v, x \rangle \leq 1 \text{ for all } x \in Q\}$$
>
> I'll work through this proof step by step to show that for any set $Q \subset \mathbb{E}$ containing the origin, the polar set Q° equals $\{v \in \mathbb{E} : \langle v, x\rangle \leq 1 \text{ for all } x \in Q\}$.
>
> First, let's recall that the polar set Q° of a set Q is defined as:
>
> $$Q^\circ = \{v \in \mathbb{E} : \langle v, x\rangle \leq 1 \text{ for all } x \in Q\}$$
>
> So we need to verify that this definition matches the set on the right-hand side of the equation. But since they're literally the same expression, this equality is trivially true by definition.

Here, Claude AI uses faulty circular reasoning, which serves as its mistake.

10. **Exercise 2.33**: Show that any convex set $Q \subset \mathbb{E}$ and a point $\bar{x} \in Q$, the equality:

$$T_Q(\bar{x}) = \operatorname{cl} \mathbb{R}_+(Q - \bar{x})$$

To prove that $T_Q(\bar{x}) = \operatorname{cl} \mathbb{R}_+(Q - \bar{x})$, we will use Double Inclusion.

First, we will show that $T_Q(\bar{x}) \subseteq \operatorname{cl} \mathbb{R}_+(Q - \bar{x})$. Let $v \in T_Q(\bar{x})$. Then, by definition, $\exists x_i \in Q$, $\tau_i \searrow 0$ such that $\lim_{i\to\infty} \tau_i^{-1}(x_i - \bar{x}) = v$. Define $y_i = \tau_i^{-1}(x_i - \bar{x})$, where $y_i \in \mathbb{R}_+(Q - \bar{x})$. Then, $v = \lim_{i\to\infty} y_i$ and so $v \in \operatorname{cl} \mathbb{R}_+(Q - \bar{x})$. Thus, we have: $T_Q(\bar{x}) \subseteq \operatorname{cl} \mathbb{R}_+(Q - \bar{x})$.

Next, we will show $\operatorname{cl} \mathbb{R}_+(Q-\bar{x}) \subseteq T_Q(\bar{x})$. Let $y \in \mathbb{R}_+(Q-\bar{x})$. That means $y = \lambda(x-\bar{x})$ for some $x \in Q$, where $\lambda \geq 0$. Let's define $x_i = \bar{x} + \lambda_i(x - \bar{x})$ with $\lambda_i \searrow 0$. We have $x_i \in Q$ because Q is convex with $x \in Q$ and $\bar{x} \in Q$, so the convex combination is in Q; $\tau_i = \lambda_i \searrow 0$; and $\tau_i^{-1}(x_i - \bar{x}) = \tau_i^{-1}(\lambda_i(x - \bar{x}))$, which is the same as $\frac{\lambda_i}{\tau_i}(x-\bar{x})$. We recall by the definition of a cone that $\lambda K \subset K$ holds for any $\lambda \geq 0$. Then, multiplying by $\lambda \geq 0$, we have:

$$\begin{aligned} y_i &= \lambda \cdot \tau_i^{-1}(x_i - \bar{x}) && \text{(By Homogeneity)} \\ &= \lambda(x - \bar{x}) && \text{(By Homogeneity)} \\ &= y && \text{(How we defined } y\text{)} \end{aligned}$$

We can state that $y = \lim_{i\to\infty} y_i$. And we know that $\lim_{i\to\infty} y_i \in T_Q(\bar{x})$. Thus, $y \in T_Q(\bar{x})$.
Therefore, $\mathbb{R}_+(Q - \bar{x}) \subseteq T_Q(\bar{x})$. Taking closures on both sides, $\operatorname{cl} \mathbb{R}_+(Q - \bar{x}) \subseteq \operatorname{cl} T_Q(\bar{x})$. We note that $\operatorname{cl} T_Q(\bar{x}) = T_Q(\bar{x})$ because $T_Q(\bar{x})$ is defined as a set of limit points and is therefore closed by definition. Thus, $\operatorname{cl} \mathbb{R}_+(Q - \bar{x}) \subseteq T_Q(\bar{x})$.

Because $T_Q(\bar{x}) \subseteq \operatorname{cl} \mathbb{R}_+(Q - \bar{x})$ and $\operatorname{cl} \mathbb{R}_+(Q - \bar{x}) \subseteq T_Q(\bar{x})$, we have that $T_Q(\bar{x}) = \operatorname{cl} \mathbb{R}_+(Q - \bar{x})$.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

We want to show:

$$T_Q(\bar{x}) = \operatorname{cl} \mathbb{R}_+(Q - \bar{x}).$$

First Inclusion:

Let $v \in T_Q(\bar{x})$. Then by definition, there exist sequences $x_i \in Q$, $\tau_i \searrow 0$, such that

$$\lim_{i \to \infty} \tau_i^{-1}(x_i - \bar{x}) = v.$$

Define $y_i = \tau_i^{-1}(x_i - \bar{x})$, then clearly $y_i \in Q - \bar{x}$. Therefore, $y_i \in \mathbb{R}_+(Q - \bar{x})$, and since $v = \lim y_i$, we conclude:

$$v \in \operatorname{cl} \mathbb{R}_+(Q - \bar{x}),$$

so

$$T_Q(\bar{x}) \subseteq \operatorname{cl} \mathbb{R}_+(Q - \bar{x}).$$

Second Inclusion:

Let $y \in \mathbb{R}_+(Q - \bar{x})$. Then there exists $x \in Q$, $\lambda > 0$, such that

$$y = \lambda(x - \bar{x}).$$

Define $x_i = \bar{x} + \lambda(x - \bar{x})$. Since $\bar{x}, x \in Q$ and Q is convex, $x_i \in Q$. Let $\tau_i = 1/i \to 0$. Then,

$$\tau_i^{-1}(x_i - \bar{x}) = \tau_i^{-1}\lambda(x - \bar{x}) = i\lambda(x - \bar{x}) \to \infty.$$

But since y is fixed and equals $\lambda(x - \bar{x})$, this contradicts the idea that the sequence converges to y. So instead, define $x_i = \bar{x} + \lambda_i(x - \bar{x})$ with $\lambda_i \searrow 0$, and take $\tau_i = \lambda_i$, then

$$\tau_i^{-1}(x_i - \bar{x}) = \tau_i^{-1}\lambda_i(x - \bar{x}) = x - \bar{x} \to y.$$

Thus, $y \in T_Q(\bar{x})$, and therefore:

$$\mathbb{R}_+(Q - \bar{x}) \subseteq T_Q(\bar{x}).$$

Taking closures:

$$\operatorname{cl} \mathbb{R}_+(Q - \bar{x}) \subseteq T_Q(\bar{x}).$$

Conclusion:

Hence, we conclude:

$$T_Q(\bar{x}) = \operatorname{cl} \mathbb{R}_+(Q - \bar{x}).$$

Here, ChatGPT makes an error by stating $y_i \in Q - \bar{x}$ means $y_i \in \mathbb{R}_+(Q - \bar{x})$. This isn't correct and is therefore a mistake.

11. **Exercise 2.36**: Show for any convex set $Q \subset \mathbb{E}$ and a point $\bar{x} \in Q$, the equality

$$N_Q(\bar{x}) = \{v \in \mathbb{E} : \langle v, x - \bar{x} \rangle \leq 0 \text{ for all } x \in Q\}$$

From **Lemma 2.35**, we know

$$N_Q(\bar{x}) = (T_Q(\bar{x}))^\circ$$

For a convex set Q, the tangent cone at $\bar{x} \in Q$ is:

$$T_Q(\bar{x}) = \operatorname{cl}\{\lambda(x - \bar{x}) : \lambda \geq 0,\ x \in Q\}$$

We see that this cone consists of all directions pointing into Q from $\bar{x}$. This is basically applying the results from **Exercise 2.33**.

By the definition of the polar cone, we have

$$(T_Q(\bar{x}))^\circ = \{v \in \mathbb{E} : \langle v, d \rangle \leq 0 \text{ for all } d \in T_Q(\bar{x})\}$$

Now, we apply the definition of the polar cone:

$$(T_Q(\bar{x}))^\circ = \{v \in \mathbb{E} : \langle v, d \rangle \leq 0 \text{ for all } d \in T_Q(\bar{x})\}$$

Note that every $d \in T_Q(\bar{x})$ is a limit of vectors of the form $\lambda(x - \bar{x})$ with $\lambda \geq 0$ and $x \in Q$ from how we showcased and described the tangent cone. We have that $\langle v, d \rangle \leq 0$ for all such d as $\langle v, x - \bar{x} \rangle \leq 0$ for all $x \in Q$. Thus, we have shown that

$$(T_Q(\bar{x}))^\circ = \{v \in \mathbb{E} : \langle v, x - \bar{x} \rangle \leq 0 \text{ for all } x \in Q\}$$

And because $N_Q(\bar{x}) = (T_Q(\bar{x}))^\circ$,

$$N_Q(\bar{x}) = \{v \in \mathbb{E} : \langle v, x - \bar{x} \rangle \leq 0 \text{ for all } x \in Q\}$$

Acknowledgment: For this question, I got help from Shawn.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

> By the definition of the polar cone, we have:
>
> $$(T_Q(\bar{x}))^\circ = \{v \in \mathbb{E} : \langle v, d \rangle \leq 0 \text{ for all } d \in T_Q(\bar{x})\}.$$
>
> Note that every $d \in T_Q(\bar{x})$ is a limit of vectors of the form $\lambda(x - \bar{x})$ with $\lambda \leq 0$ and $x \in Q$, based on how we characterized the tangent cone. Thus, if $\langle v, x - \bar{x} \rangle \leq 0$ for all $x \in Q$, then $\langle v, d \rangle \leq 0$ for all $d \in T_Q(\bar{x})$. Hence,
>
> $$(T_Q(\bar{x}))^\circ = \{v \in \mathbb{E} : \langle v, x - \bar{x} \rangle \leq 0 \text{ for all } x \in Q\},$$
>
> and therefore:
>
> $$N_Q(\bar{x}) = \{v \in \mathbb{E} : \langle v, x - \bar{x} \rangle \leq 0 \text{ for all } x \in Q\}.$$

The problem here is that ChatGPT discusses every d as a limit of vectors of the form $\lambda(x - \bar{x})$ with $\lambda \leq 0$; this is a mistake as it should be $\lambda \geq 0$.

12. **Exercise 2.37**: Prove that the following properties are equivalent for any nonempty, closed, convex set Q, a point $\bar{x} \in Q$, and a vector $v \in \mathbb{E}$.

    1. $v \in N_Q(\bar{x})$
    2. $\bar{x} \in \operatorname*{arg\,max}_{x \in Q} \langle v, x \rangle$
    3. $\text{proj}_Q(\bar{x} + \lambda v) = \bar{x}$ for all $\lambda \geq 0$
    4. $\text{proj}_Q(\bar{x} + \bar{\lambda} v) = \bar{x}$ for some $\bar{\lambda} > 0$

    We will show that the four statements are equivalent.

    Statement 1 implies Statement 2:
    We will show that if $v \in N_Q(\bar{x})$, then $\bar{x} \in \operatorname*{arg\,max}_{x \in Q} \langle v, x \rangle$. Let $v \in N_Q(\bar{x})$. Then, $\langle v, x - \bar{x} \rangle \leq 0$ for all $x \in Q$. Thus, $\langle v, x \rangle \leq \langle v, \bar{x} \rangle$ for all $x \in Q$. This gives us by definition, $\bar{x} \in \operatorname*{arg\,max}_{x \in Q} \langle v, x \rangle$.
    Thus, if $v \in N_Q(\bar{x})$, then $\bar{x} \in \operatorname*{arg\,max}_{x \in Q} \langle v, x \rangle$.

    Statement 2 implies Statement 3:
    We will show that if $\bar{x} \in \operatorname*{arg\,max}_{x \in Q} \langle v, x \rangle$, then $\text{proj}_Q(\bar{x} + \lambda v) = \bar{x}$ for all $\lambda \geq 0$. Consider any $\lambda \geq 0$. Define the projection: $y_\lambda = \text{proj}_Q(\bar{x} + \lambda v)$. By definition of projection, we know that $y_\lambda \in Q$ is the unique point that minimizes the distance to $\bar{x} + \lambda v$. Thus, by properties of projection onto a convex set, we have the optimality condition:

    $$\langle (\bar{x} + \lambda v) - y_\lambda, x - y_\lambda \rangle \leq 0\,, \quad \forall x \in Q$$

    In particular, set $x = \bar{x}$ since $\bar{x} \in Q$:

    $$\langle (\bar{x} + \lambda v) - y_\lambda, \bar{x} - y_\lambda \rangle \leq 0$$

    $$\langle \bar{x} - y_\lambda, \bar{x} - y_\lambda \rangle + \lambda \langle v, \bar{x} - y_\lambda \rangle \leq 0$$
    $$\|\bar{x} - y_\lambda\|^2 + \lambda \langle v, \bar{x} - y_\lambda \rangle \leq 0$$

    By definition of $\operatorname*{arg\,max}_{x \in Q} \langle v, x \rangle$, $\bar{x}$ maximizes over $\langle v, x \rangle$ over $x \in Q$. Thus, for all $x \in Q$:

    $$\langle v, x - \bar{x} \rangle \leq 0$$

    $$\langle v, \bar{x} - x \rangle \geq 0$$

    Since $y_\lambda \in Q$, we specifically have $\langle v, \bar{x} - y_\lambda \rangle \geq 0$. Thus, because $\lambda \geq 0$, $\lambda \langle v, \bar{x} - y_\lambda \rangle \geq 0$. Furthermore, $\|\bar{x} - y_\lambda\|^2 \geq 0$ by definition of the norm. Combining our two inequalities, $\lambda \langle v, \bar{x} - y_\lambda \rangle + \|\bar{x} - y_\lambda\|^2 \geq 0$. We also have that $\lambda \langle v, \bar{x} - y_\lambda \rangle + \|\bar{x} - y_\lambda\|^2 \leq 0$. The only way that both $\lambda \langle v, \bar{x} - y_\lambda \rangle + \|\bar{x} - y_\lambda\|^2 \geq 0$ and $\lambda \langle v, \bar{x} - y_\lambda \rangle + \|\bar{x} - y_\lambda\|^2 \leq 0$ is when $\lambda \langle v, \bar{x} - y_\lambda \rangle + \|\bar{x} - y_\lambda\|^2 = 0$. Additionally, since $\lambda \langle v, \bar{x} - y_\lambda \rangle \geq 0$ and $\|\bar{x} - y_\lambda\|^2 \geq 0$, that means that both $\lambda \langle v, \bar{x} - y_\lambda \rangle = 0$ and $\|\bar{x} - y_\lambda\|^2 = 0$. Because $\|\bar{x} - y_\lambda\|^2 = 0$, $y_\lambda = \bar{x}$. That means we have $\text{proj}_Q(\bar{x} + \lambda v) = \bar{x}$ for all $\lambda \geq 0$.
    Thus, if $\bar{x} \in \operatorname*{arg\,max}_{x \in Q} \langle v, x \rangle$, then $\text{proj}_Q(\bar{x} + \lambda v) = \bar{x}$ for all $\lambda \geq 0$.

    Statement 3 implies Statement 4:
    If $\text{proj}_Q(\bar{x} + \lambda v) = \bar{x}$ for all $\lambda \geq 0$, it immediately follows that $\text{proj}_Q(\bar{x} + \bar{\lambda} v) = \bar{x}$ for some $\bar{\lambda} > 0$ as any value where $\lambda > 0$ will satisfy $\text{proj}_Q(\bar{x} + \bar{\lambda} v) = \bar{x}$ for some $\bar{\lambda} > 0$.

    Statement 4 implies Statement 1:
    We will show that if $\text{proj}_Q(\bar{x} + \bar{\lambda} v) = \bar{x}$ for some $\bar{\lambda} > 0$, then $v \in N_Q(\bar{x})$. Let $\lambda > 0$ as we know

this has to be the case for some λ. If $\operatorname{proj}_Q(\bar{x} + \lambda v) = \bar{x}$, then by the projection optimality condition, $\langle \bar{x} + \lambda v - \bar{x}, x - \bar{x} \rangle \leq 0$ for all $x \in Q$. Simplifying, we get $\langle \lambda v, x - \bar{x} \rangle \leq 0$ for all $x \in Q$. Dividing by $\lambda > 0$, we obtain $\langle v, x - \bar{x} \rangle \leq 0$ for all $x \in Q$. This is precisely the definition of $v \in N_Q(\bar{x})$. Thus, if $\operatorname{proj}_Q(\bar{x} + \bar{\lambda} v) = \bar{x}$ for some $\bar{\lambda} > 0$, then $v \in N_Q(\bar{x})$.

Because Statement 1 implies Statement 2, Statement 2 implies Statement 3, Statement 3 implies Statement 4, and Statement 4 implies Statement 1, we have that from any statement, we can arrive at another statement meaning that all four properties are equivalent for any nonempty, closed, convex set Q, a point $\bar{x} \in Q$, and a vector $v \in \mathbb{E}$.

Acknowledgment: For this question, I got help from Batool and Shawn.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

> We will show that if $\bar{x} \in \arg\max_{x \in Q} \langle v, x \rangle$, then $\operatorname{proj}_Q(\bar{x} + \lambda v) = \bar{x}$ for all $\lambda \geq 0$. Since $\bar{x}$ maximizes the inner product $\langle v, x \rangle$ over Q, it lies at a point on the **interior** of the feasible region Q, where the supporting hyperplane orthogonal to v touches the set.
>
> Thus, any movement in the direction of v will move outside the set Q, and the projection of $\bar{x} + \lambda v$ will remain at $\bar{x}$.

Here, the error is that ChatGPT states that since $\bar{x}$ maximizes the inner product $\langle v, x \rangle$ over Q, it lies at a point on the interior of the feasible region Q. This is a mistake as it is untrue. In fact, since $\bar{x}$ maximizes the inner product $\langle v, x \rangle$ over Q, it lies at a point on the relative boundary of the feasible region Q. Furthermore, $\operatorname{rb} Q = (\operatorname{cl} Q) \setminus (\operatorname{ri} Q)$.

13. **Exercise 2.39**: Show for any convex set Q and a point $x \in Q$, the equivalence

$$x \in \operatorname{int} Q \quad \text{if and only if } N_Q(x) = \{0\}$$

We will do this Proof by Biconditional:

We will prove that if $x \in \operatorname{int} Q$, then $N_Q(x) = \{0\}$. Let $x \in \operatorname{int} Q$. Then, there exists a radius r such that $B_r(x) \subseteq Q$. Now, suppose $v \in N_Q(x)$. Then, by definition, $\langle v, y - x\rangle \leq 0$ for all $y \in Q$. In particular, for any $u \in \mathbb{E}$ with $\|u\| = 1$, we can choose $y = x + \delta u$, such that $y \in Q$ for sufficiently small $\delta > 0$. Additionally, since $x \in \operatorname{int} Q$, the whole ball, $B_r(x)$, is contained in Q. Then:

$$\begin{aligned}\langle v, x + \delta u - x\rangle &= \langle v, \delta u\rangle \\ &= \delta\langle v, u\rangle \\ &\leq 0\end{aligned}$$

$$\langle v, u\rangle \leq 0$$

But, we could also choose $y = x - \delta u$, such that $y \in Q$, and similarly get: $\langle v, -u\rangle \leq 0$, which means $\langle v, u\rangle \geq 0$.

Now, we combine both inequalities:

$$\langle v, u\rangle \leq 0 \text{ and } \langle v, u\rangle \geq 0$$

This means:

$$\langle v, u\rangle = 0$$

Since this is true for every unit vector $u \in \mathbb{E}$, we conclude: $v = 0$. We notice that $0 \in N_Q(x)$ and only 0 satisfies both $\langle v, u\rangle \leq 0$ and $\langle v, u\rangle \geq 0$. Thus, $N_Q(x) = \{0\}$. So, if $x \in \operatorname{int} Q$, then $N_Q(x) = \{0\}$.

We will now prove that if $N_Q(x) = \{0\}$, then $x \in \operatorname{int} Q$. We will actually do this by using the contrapositive statement of if $N_Q(x) = \{0\}$, then $x \in \operatorname{int} Q$. So, we will show if $x \notin \operatorname{int} Q$, then $N_Q(x) \neq \{0\}$. Let $x \notin \operatorname{int} Q$ and so x is on the boundary of Q. By Strict Separation, we know that there is some $v \in \mathbb{E}$ and $b \in \mathbb{R}$ such that

$$\langle v, y\rangle \leq b < \langle v, x\rangle \text{ for all } y \in Q$$

That is

$$\langle v, y - x\rangle \leq 0 \text{ for all } y \in Q$$

That means $v \in N_Q(x)$ and by Strict Separation we know that this v is a nonzero vector. This means $N_Q(x) \neq \{0\}$. Thus, if $x \notin \operatorname{int} Q$, then $N_Q(x) \neq 0$. That means if $N_Q(x) = \{0\}$, then $x \in \operatorname{int} Q$ by definition of contrapositive statements.

Because if $x \in \operatorname{int} Q$, then $N_Q(x) = \{0\}$ and $N_Q(x) = \{0\}$, then $x \in \operatorname{int} Q$, we have that
$x \in \operatorname{int} Q$ if and only if $N_Q(x) = \{0\}$.

Relationship between normal vectors and supporting halfspaces: Given a convex set $Q \subseteq \mathbb{E}$, a supporting halfspace at a boundary point $x \in \operatorname{bd} Q$ is defined as a halfspace $H \subseteq \mathbb{E}$ such that $x \in \operatorname{bd} H$ and $Q \subseteq H$. If $v \in N_Q(x)$ is a normal vector at x, then the halfspace defined by $H = \{y \in \mathbb{E} : \langle v, y - x \rangle \leq 0\}$ is a supporting halfspace for Q at x. Indeed, this halfspace satisfies $x \in \operatorname{bd} H$ because $\langle v, x - x \rangle = 0$ and $Q \subseteq H$ because by definition of the normal cone $N_Q(x)$, we have $\langle v, y - x \rangle \leq 0$ for all $y \in Q$. Conversely, every supporting halfspace at a point $x \in \operatorname{bd} Q$ is associated with some nonzero normal vector $v \in N_Q(x)$ by Strict Separation whose corresponding halfspace boundary exactly defines this supporting halfspace. In conclusion, normal vectors $v \in N_Q(x)$ characterizes the orientation of supporting halfspaces to a convex set Q at its boundary points.

Acknowledgment: For this question, I got help from Shawn.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Suppose $x \in \operatorname{int} Q$. Then by definition, the point x is not in Q, so clearly the normal cone at x must be empty:

$$N_Q(x) = \emptyset$$

Since the only set containing no vectors is the empty set, and $\{0\} \neq \emptyset$, we conclude:

$$N_Q(x) = \{0\} \Rightarrow x \notin Q$$

Therefore, the claim is proved.

Here, ChatGPT makes a mistake by stating that $N_Q(x) = \emptyset$, which is untrue. Furthermore, if $N_Q(x) = \emptyset$, $0 \notin N_Q(x)$ by definition of the empty set. Rather, $N_Q(x) \neq \emptyset$ and $0 \in N_Q(x)$.